Introduction to Databases

CSE 341
Introduction to Database Systems
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CSE 341

Introduction to Databases

- This course offers an overview of database systems, including the essentials of database design, modeling, and architecture with practical applications. The course begins with foundational concepts including entity-relationship models, the relational model, and SQL.

Today

- The importance of database systems and some common uses of database systems.
- File-based systems: characteristics, problems.
- “database.”
- “database management system” (DBMS).

Database Applications in Daily Life

- Businesses: Banking, Insurance, Retail, Transportation, Healthcare, Manufacturing
- Service Industries: Financial, Real-estate, Legal, Electronic Commerce, Small businesses
- Education : Resources for content and Delivery
- More recently: Social Networks, Environmental and Scientific Applications, Medicine and Genetics
- Personalized Applications: based on smart mobile devices

File-Based Systems

- A tradition to introduce the database system with a review of its predecessor, the file-based system.

- File-based systems were an early attempt to computerize the manual filing system.

File-Based Systems

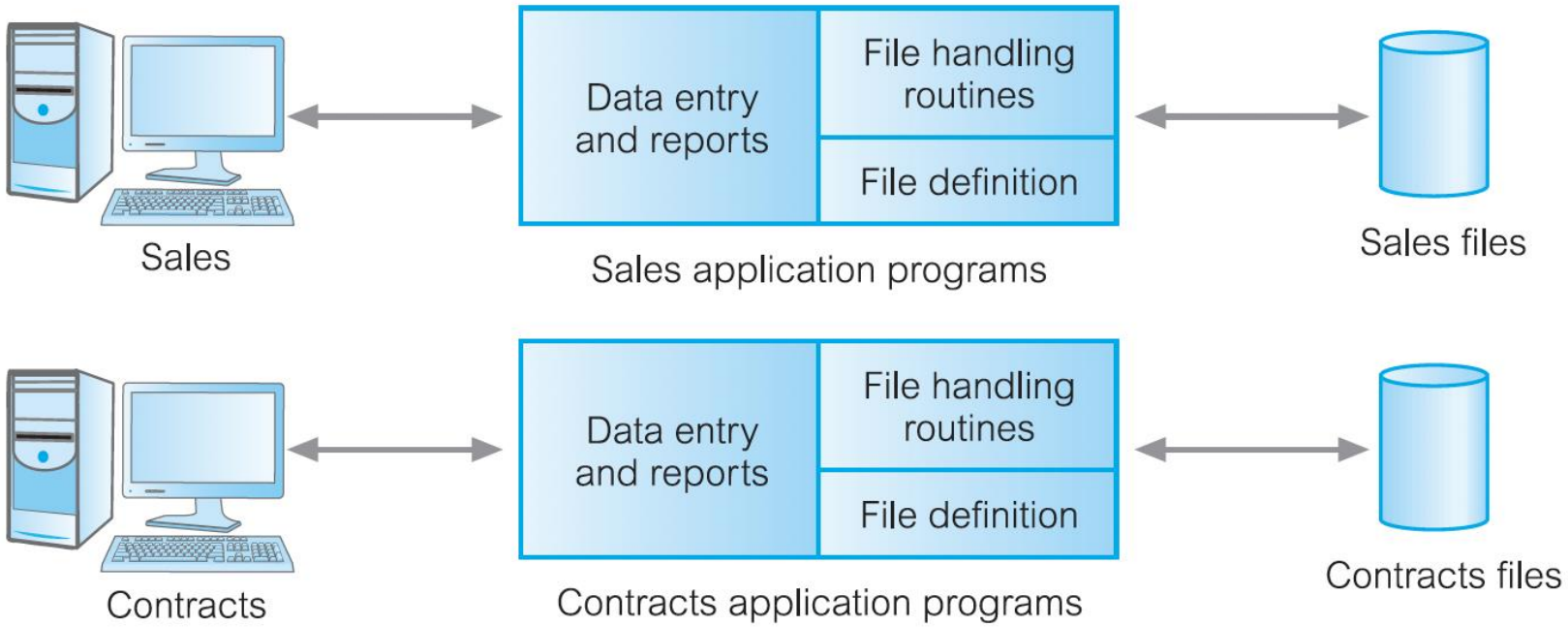


Figure 1.5 File-based processing.

Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)
PropertyForRent (propertyNo, street, city, postcode, rent)
Client (clientNo, fName, lName, address, telNo)

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead Rd	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Client

clientNo	fName	lName	address	telNo	prefType	maxRent
CR76	John	Kay	56 High St, London SW1 4EH	0207-774-5632	Flat	425
CR56	Aline	Stewart	64 Fern Dr, Glasgow G42 0BL	0141-848-1825	Flat	350
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178	House	750
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720	Flat	600

Figure 1.2 The PropertyForRent, PrivateOwner, and Client files used by Sales.

Figure 1.4
The Lease, PropertyForRent, and Client files used by the Contracts Department.

Lease

leaseNo	propertyNo	clientNo	rent	payment Method	deposit	paid	rentStart	rentFinish	duration
10024	PA14	CR62	650	Visa	1300	Y	1-Jun-13	31-May-14	12
10075	PL94	CR76	400	Cash	800	N	1-Aug-13	31-Jan-14	6
10012	PG21	CR74	600	Cheque	1200	Y	1-Jul-13	30-Jun-14	12

PropertyForRent

propertyNo	street	city	postcode	rent
PA14	16 Holhead	Aberdeen	AB7 5SU	650
PL94	6 Argyll St	London	NW2	400
PG21	18 Dale Rd	Glasgow	G12	600

Client

clientNo	fName	lName	address	telNo
CR76	John	Kay	56 High St, London SW1 4EH	0171-774-5632
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720

Limitations of File-Based Systems

Separation and isolation of data

- Each program maintains its own set of data.
- Users of one program may be unaware of potentially useful data held by other programs.

Duplication of data

- Same data is held by different programs.
- Wasted space and potentially different values and/or different formats for the same item.

Limitations of File-Based Systems

Data dependence

- File structure is defined in the program code.

Incompatible file formats

- Programs are written in different languages, and so cannot easily access each other's files.

Fixed Queries

- Programs are written to satisfy particular functions.
- Any new requirement needs a new program.

(1) The definition of the data is embedded in the application programs, rather than being stored separately and independently.

(2) There is no control over the access and manipulation of data beyond that imposed by the application programs.

- As a result, a new approach was required.
- What emerged were the database and the Database Management System (DBMS).

What is a Database?

A shared collection of logically related data and its description, designed to meet the information needs of an organization.

- a single, possibly large repository of data that can be used simultaneously by many departments and users.
- a shared corporate resource.
- The database holds not only the organization's operational data, but also a description of this data ➔ *a self-describing collection of integrated records*.

What is a DBMS?

A software system that enables users to define, create, maintain, and control access to the database.

- the software that interacts with the users' application programs and the database.

Core Functionalities of a DBMS

- Define the database, usually through a **Data Definition Language (DDL)**. The DDL allows users to specify the data types and structures and the constraints on the data to be stored in the database.
- Insert, update, delete, and retrieve data from the database, usually through a **Data Manipulation Language (DML)**. The most common query language is the **Structured Query Language** (**SQL**, pronounced "S-Q-L", or sometimes "See-Quel").
- Provide controlled access to the database (e.g. security system, concurrency control system, recovery system, etc.)

Components of DBMS Environment

Hardware

- Can range from a PC to a network of computers.

Software

- DBMS, operating system, network software (if necessary) and also the application programs.

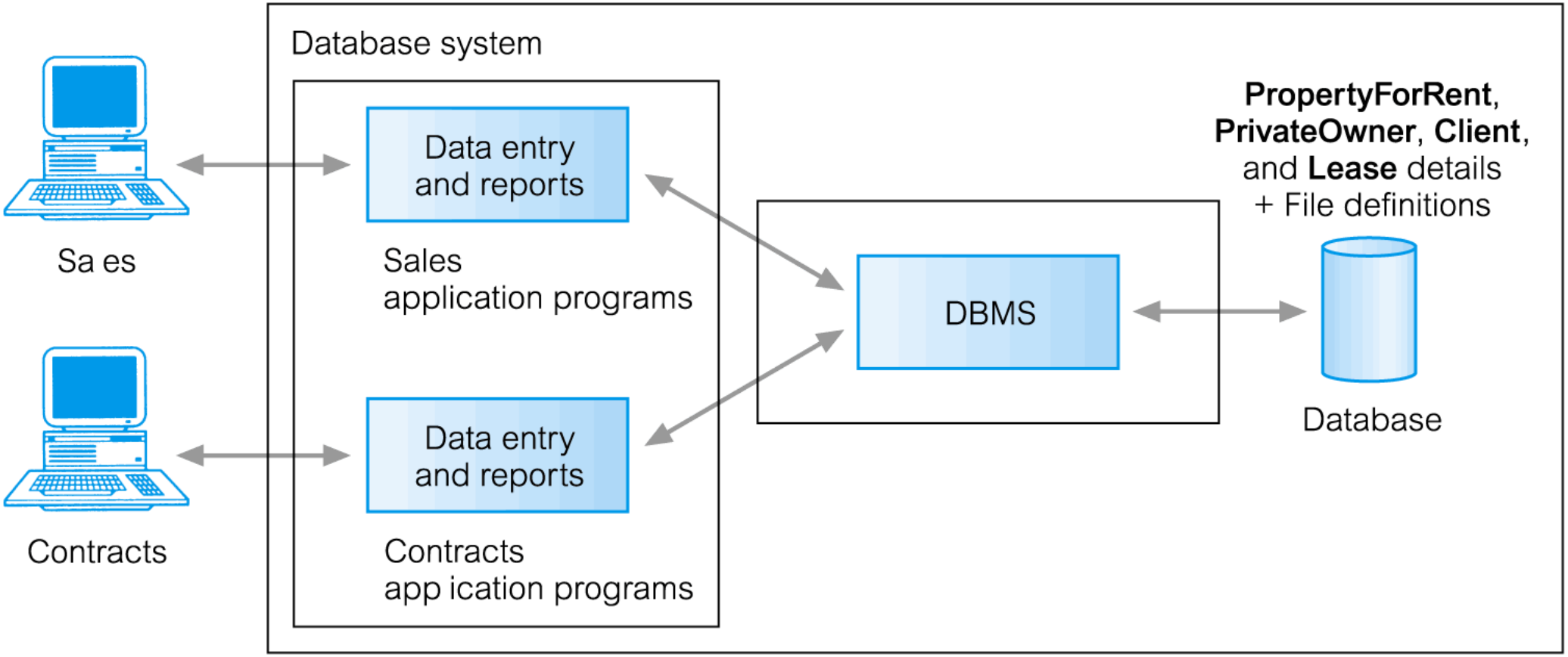
Data

- Used by the organization and a description of this data called the schema.

Procedures

- Instructions and rules that should be applied to the design and use of the database and DBMS.

People



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Simplified Database System Environment

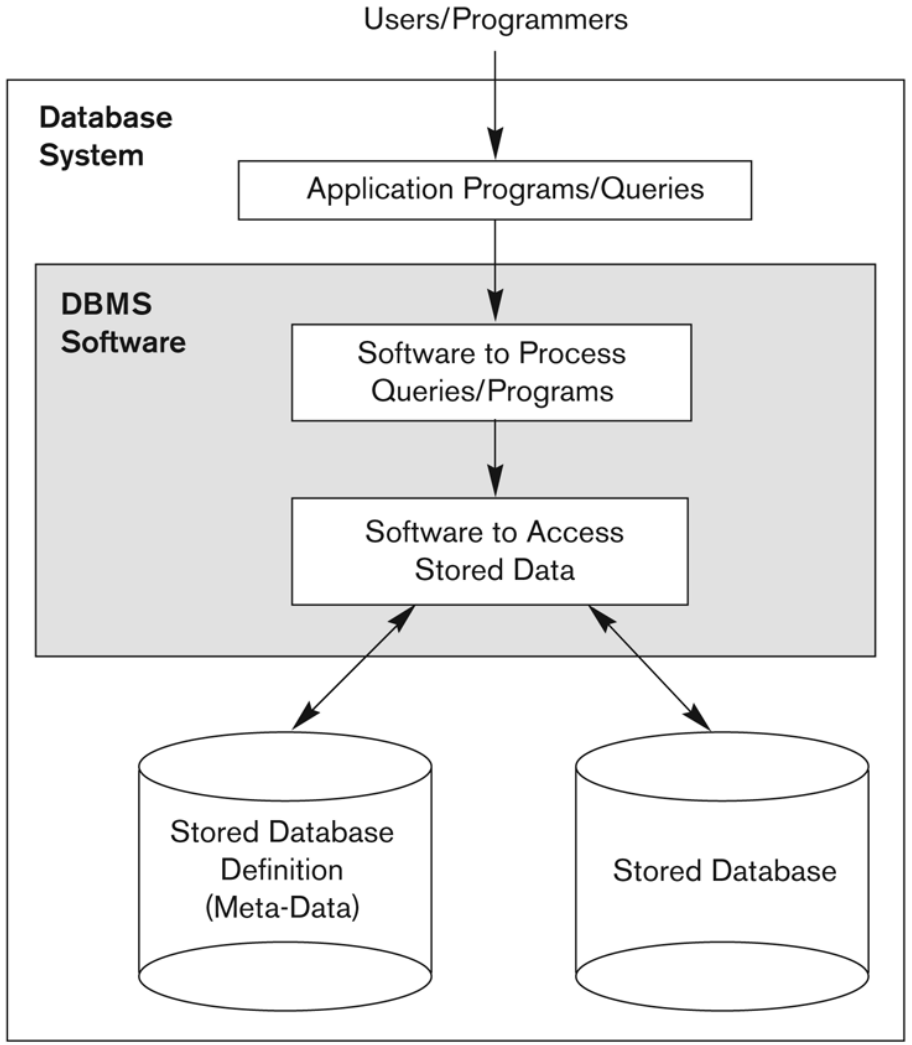

Figure 1.1
A simplified database system environment.

Database Users and Their Role

- Data Administrator (DA): responsible for the management of the data resource, including database planning; development and maintenance of standards, policies and procedures
- Database Administrator (DBA): responsible for the physical realization of the database
- Database Designers
  - Logical: identifying the data (that is, the entities and attributes), the relationships between the data, and the constraints on the data
  - Physical: how the logical database design is to be physically realized
- Application Programmers
- End Users (naive and sophisticated)

History of Database Systems

- First-generation
  - Hierarchical and Network
- Second generation
  - Relational
- Third generation
  - Object-Relational
  - Object-Oriented

Advantages of DBMSs

- Control of data redundancy
- Data consistency
- More information from the same amount of data
- Sharing of data
- Improved data integrity
- Improved security
- Enforcement of standards
- Economy of scale

- And more

Challanges of DBMSs

- Complexity
- Size
- Cost of DBMS
- Additional hardware costs
- Cost of conversion
- Performance
- Higher impact of a failure